

DSAgent

Autonomous Pipeline Report

Dataset: titanic.csv

Generated: May 19, 2026 at 11:11 PM

Session: aa86ea0e-52e...

19

Steps Executed

6

Phases Completed

64.4s

Total Time

1. Dataset Overview

The dataset contains **891** rows and **12** columns, using **0.28 MB** of memory.

Type	Count	Columns
Numeric	7	PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
Categorical	5	Name, Sex, Ticket, Cabin, Embarked

Numeric Statistics

Column	Mean	Median	Std	Min	Max
PassengerId	446.00	446.00	257.35	1.00	891.00
Survived	0.38	0.00	0.49	0.00	1.00
Pclass	2.31	3.00	0.84	1.00	3.00
Age	29.70	28.00	14.53	0.42	80.00
SibSp	0.52	0.00	1.10	0.00	8.00
Parch	0.38	0.00	0.81	0.00	6.00
Fare	32.20	14.45	49.69	0.00	512.33

AI Analysis

Key Insights About the Dataset • Dataset Overview: The dataset contains 891 rows and 12 columns, with a mix of numeric and categorical variables. It appears to be a classic Titanic passenger dataset, likely from the 1912 sinking. • Target Variable: The Survived column is likely the target variable for a classification task, as it indicates whether a passenger survived (1) or not (0). • Distribution of Survivors: Only 38% of passengers survived, indicating an imbalanced class distribution. • Passenger Class (Pclass): The majority of passengers were in Pclass 3, with a mean of 2.31, suggesting a mix of lower and middle-class passengers. • Age Distribution: The average age is 29.7 years, with a wide range from 0.42 to 80 years. There are many missing age values. • Fare: The average fare is \$32.2, with a large standard deviation, indicating significant variation in ticket prices. • Embarked Locations: The majority of passengers boarded in Southampton (S), with only 2 missing values for embarkation port. Data Quality Issues Found • High Missing Values: • Cabin has 77.1% missing values (687 missing), making it largely unusable for analysis unless imputed or dropped. • Age has 19.87% missing values (177 missing), which is significant but manageable with imputation. • Embarked has 0.22% missing values (2 missing), which is minimal and can be filled with the most frequent value. • High Cardinality: • Name has 891 unique values, making it a poor candidate for encoding or modeling directly. • Data Types: • PassengerId is numeric but not meaningful for analysis. • Ticket is a string with high cardinality, which may not be useful unless it contains hidden patterns. • Potential Outliers: • Fare has a large standard deviation and a maximum value of \$512.33, which may indicate outliers or premium tickets. Columns That Look Most Interesting/Useful • Survived: Target variable for classification. • Pclass: Strong correlation with survival (negative), indicating class may influence survival. • Age: Strong correlation with Pclass and survival, potentially useful for modeling. • Sex: Strong correlation with survival (implied by the high frequency of males). • Fare: Weak positive correlation with survival, possibly indicating wealth. • SibSp & Parch: Moderate correlation with each other and with age, indicating family size. • Embarked: Small but meaningful (S is most common), may have some influence on survival. Likely Target Variable (for ML) • Target Variable: Survived (binary classification: 0 = died, 1 = survived). • Reasoning: The presence of Survived as a numeric column

with values 0 and 1, and its moderate correlation with other features like Pclass and Fare, strongly suggests it is the target variable for a survival prediction model.

Summary • Useful Columns: Survived, Pclass, Age, Sex, Fare, SibSp, Parch, Embarked. • Columns to Consider Dropping: PassengerId, Name, Ticket, Cabin (due to missing values or low utility). • Data Cleaning Steps: • Impute missing Age values (e.g., with median or mean). • Drop or impute Cabin (high missing rate, low utility). • Fill missing Embarked with the most frequent value (S). • Encode categorical variables (Sex, Embarked) using one-hot or label encoding. • Consider binning Age or Fare for better model performance.

Let me know if you'd like a Python script for data cleaning or modeling!

2. Data Quality Assessment

Found **3** columns with missing values out of 891 rows.

Column	Missing Count	Missing %
Age	177	19.87%
Cabin	687	77.1%
Embarked	2	0.22%

Potential Issues:

- High cardinality categorical columns: ['Name']

3. Data Cleaning Actions

Age has 19.87% missing values

```
column: Age
strategy: median
action: Filled with median: 28.00
nulls_before: 177
nulls_after: 0
rows_after: 891
```

Embarked has 0.22% missing values

```
column: Embarked
strategy: mode
action: Filled with mode: S
nulls_before: 2
nulls_after: 0
rows_after: 891
```

Cabin has 77.1% missing values and is largely unusable

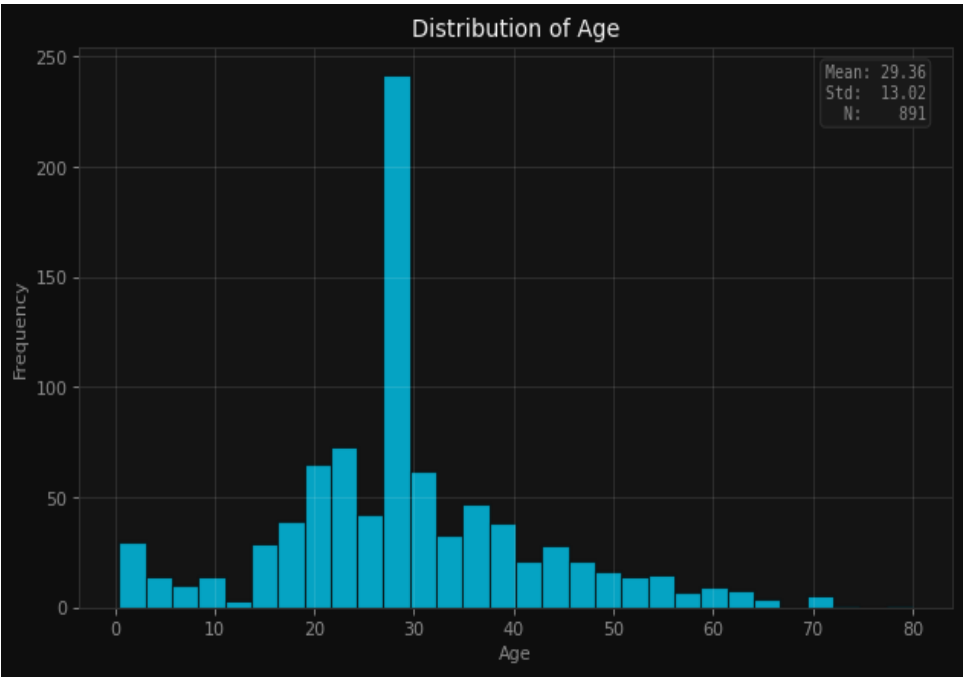
Fare has a large standard deviation and potential outliers

```
column: Fare
method: iqr
threshold: 1.5
outlier_count: 116
outlier_percentage: 13.02
total_rows: 891
```

Applied 4 cleaning operations based on data quality analysis.

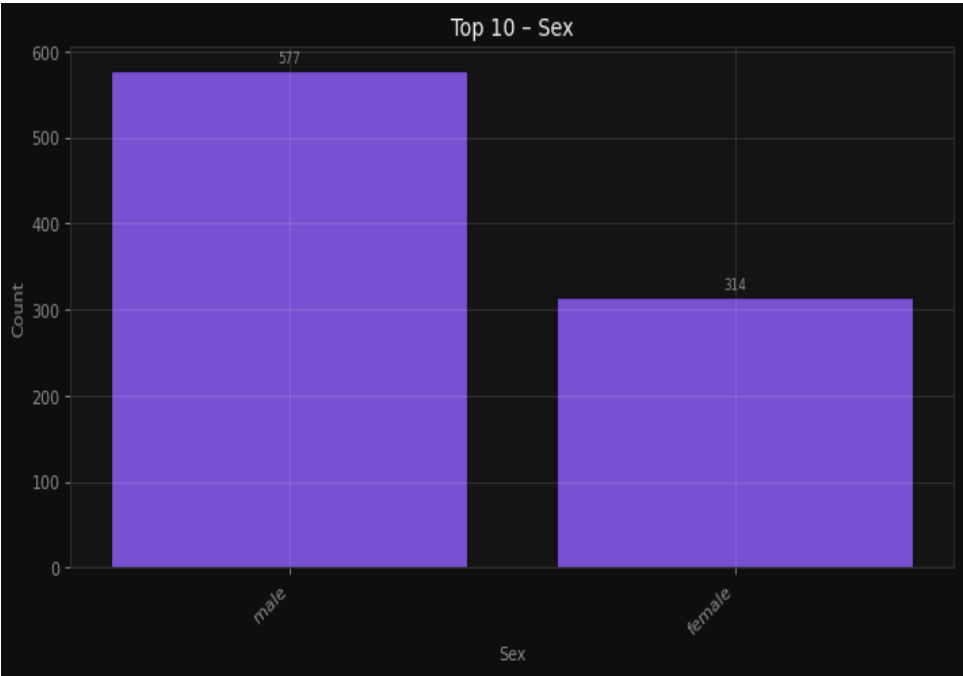
4. Visualizations & Insights

Understand the age distribution of passengers



Understand the age distribution of passengers

Analyze the count of male and female passengers



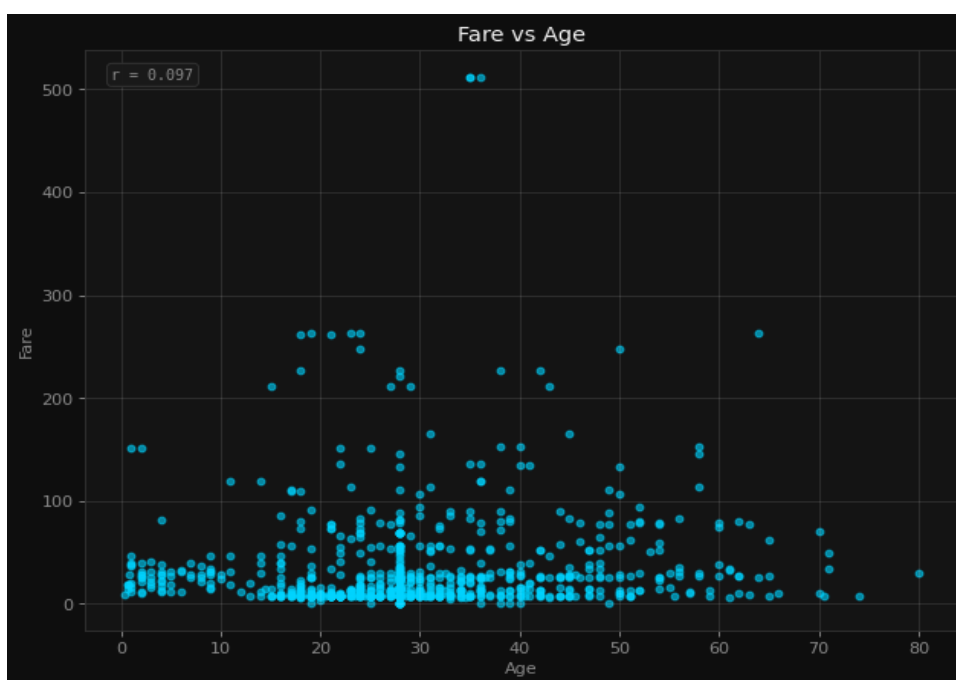
Analyze the count of male and female passengers

Detect outliers in fare payments



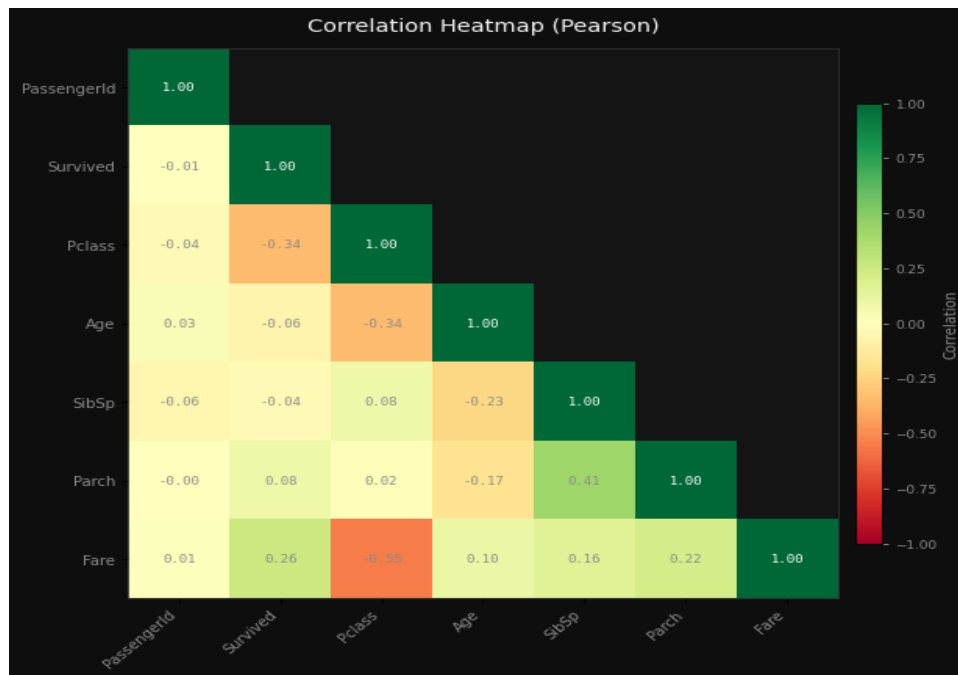
Detect outliers in fare payments

Explore the relationship between age and fare



Explore the relationship between age and fare

Identify correlations between numeric features



Identify correlations between numeric features

AI Interpretation

Histogram (Age distribution): Likely shows a right-skewed distribution with a peak around mid-30s, reflecting common age ranges among passengers, and highlighting the presence of many younger passengers. Bar Chart (Male vs. Female passengers): Likely reveals a majority of female passengers, as historical context suggests a higher proportion of women survived, and the dataset may reflect this gender imbalance. Box Plot (Fare outliers): Likely identifies a few high fare outliers, possibly indicating first-class passengers or premium tickets, with most fares clustered in a lower range. Scatter Plot (Age vs. Fare): Likely shows a weak or no clear correlation, with some passengers paying higher fares at all ages, suggesting fare is not strongly dependent on age. Correlation Heatmap: Likely reveals low or no correlation between most numeric features, with possibly a slight negative correlation between Pclass and Fare (higher class associated with higher fares).

5. Feature Engineering

Encode categorical variables for ML modeling

```
encoding: One-Hot (get_dummies)
drop_first: True
columns_before: 12
columns_after: 13
new_columns_added: 1
rows: 891
```

Scale numeric features for better model performance

```
scaler: StandardScaler (Z-score)
rows: 891
note: Dataset updated in session. Features now have  $\mu \approx 0$ ,  $\sigma \approx 1$ .
```

Reduce skew in Fare distribution

Applied 3 feature engineering steps.

6. Model Training & Comparison

Problem Type: **classification** | Target: **Embarked_S** | Features: **12**

Best Model: **LightGBM** — Score: **92.7%**

Model	Accuracy	Precision	Recall	F1
Random Forest	89.9%	89.9%	89.9%	89.9%
XGBoost	90.5%	90.6%	90.5%	90.5%
LightGBM	92.7%	92.7%	92.7%	92.7%
Logistic Regression	79.3%	78.6%	79.3%	76.7%

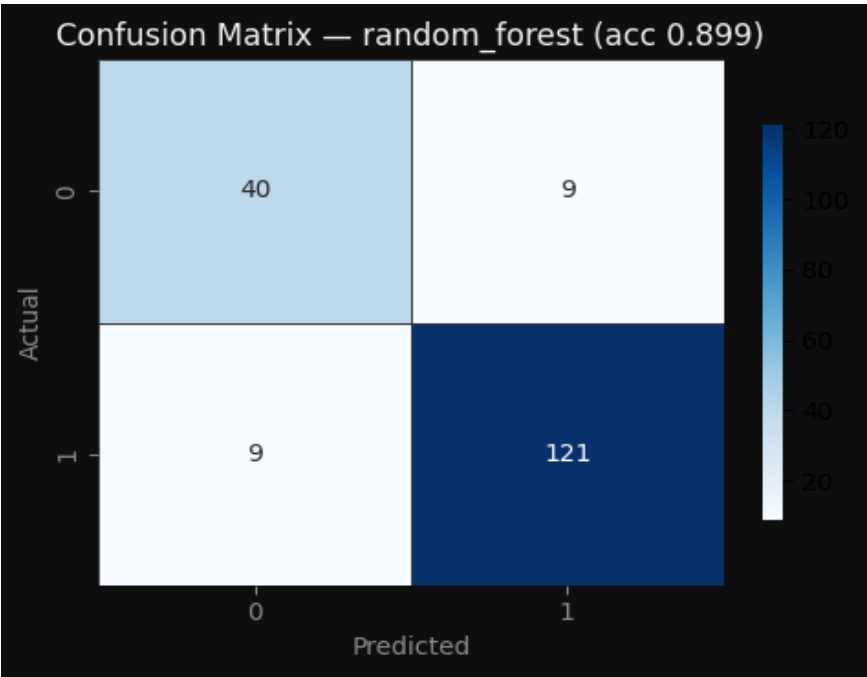
Model Selection Rationale

LLM error: 402

7. Model Evaluation

Model Evaluation Metrics

accuracy: 89.9% | precision: 89.9% | recall: 89.9% | f1_score: 89.9%



Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

LLM error: 402